

BYKOVSKII IN THE RANK-2 SEMILOCAL CASE

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Let R be a commutative ring. The rank- r *Steinberg module* $\text{St}_r(R)$ associated to R is the top reduced homology of the Tits building $T_r(R)$, the order complex of the poset of projective summands of R^r . For many rings R , including all domains and semilocal rings, this is the only nonvanishing degree of reduced homology of $T_r(R)$, which is in fact even a homotopy Cohen–Macaulay simplicial complex in these cases. When R is a domain with fraction field F , one may observe that projective summands of F^r and of R^r are in bijection, meaning that the natural inclusion $T_r(R) \rightarrow T_r(F)$ is a homeomorphism. This yields a canonical identification $\text{St}_r(R) \cong \text{St}_r(F)$.

For each R -basis $v_1, \dots, v_r$ of R^r , the barycentric subdivision of the $(r-1)$ -simplex with vertices labelled by these has facets naturally are labelled by flags of subsets of these basis elements; its boundary hence includes naturally into $T_r(R)$ and yields a $(r-2)$ -cycle giving a class $[v_1, \dots, v_r]$ in $\text{St}_r(R)$. For many R , including all semi-local rings R and Euclidean domains, these apartment classes generate.

Definition 0.1. We say that $\text{St}_r(R)$ has a *Bykovskii presentation* if apartment classes generate and give a presentation with the following relations:

- (1) $[v_1, \dots, v_r] = [av_1, \dots, v_r]$ for any $a \in R^\times$.
- (2) $[v_{\sigma 1}, \dots, v_{\sigma r}] = (-1)^{\text{sgn } \sigma} [v_1, \dots, v_r]$ for any $\sigma \in S_r$.
- (3) $[v_1, \dots, v_r] - [u_1 v_1 + u_2 v_2, v_2, \dots, v_r] + [u_1 v_1 + u_2 v_2, v_1, v_3, \dots, v_r] = 0$ for any units $u_1, u_2 \in R^\times$.

Theorem 0.2. *If R is semi-local with all residue fields of size at least 4, then $\text{St}_2(R)$ has a Bykovskii presentation.*

Proof. The method of [CP] (and [KMPW] in a more general setting) implies that the Bykovskii presentation holds if and only if the H_1 of the *partial basis complex* $B_2(R)$, whose vertices are free rank-1 summands of R^2 , and 1-simplices are pairs of summands giving a direct sum decomposition, is generated by triangles. Equivalently, the obstruction to the Bykovskii presentation holding is the first homology of the *augmented complex* $BA_2(R)$ where one glues all triangles to $B_2(R)$.

Note that two such lines give a direct sum decomposition if and only if they do so modulo each of the finitely many maximal ideals of R .

Suppose we have a cycle $v_1 v_2 \dots v_k$ of length $k \geq 4$ in $B_2(R)$. We may write it as a sum of the shorter cycles $v_1 v_2 u$, $v_2 v_3 u$, $v_3 v_4 u$, and $v_4 v_5 \dots v_k v_1 u$ so long as u forms a basis with each of v_1, v_2, v_3, v_4 . Indeed, since each residue field is size at least 4, there are at least 5 distinct lines in each residual vector space of R^2 ; by the Chinese remainder theorem, we may simply

pick u to span a line distinct from any of the residue classes of $v_1, \dots, v_4$ modulo each maximal ideal. Then descending in this fashion, we may decompose any cycle into triangles. $\square$

Remark 0.3. One expects from larger conjectures (in particular Rognes connectivity) that relations in the Steinberg module of semilocal rings are “quadratically generated” (cf. [X2, Appendix A], [KMPW, Question 5.2]) and so that Bykovskii relations should hold for (most) semilocal rings in arbitrary rank.

The residue field bound size is not sharp; $\mathbb{Z}/10 = \mathbb{F}_2 \times \mathbb{F}_5$ also has H_1 generated by triangles, for example. However, the theorem can be false with no residue field size restriction: for example, for $R = \mathbb{Z}/6$, $BA_2(R)$ is the chessboard complex $M_{3,4}$, which is a triangulation of a torus, so its H_1 is nonzero and yields relations in the Steinberg module which cannot be generated by three-term Bykovskii relations. The same failure can be lifted to the semilocal domain $\mathbb{Z}_{(6)}$: its $BA_2(\mathbb{Z}_{(6)})$ is the inflation of $BA_2(\mathbb{Z}/6)$ (since on the 1-skeletons the basis property can be detected mod 6, and then we just glue all triangles), so its H_1 is also nonzero. Concretely, for example, we may consider 4-cycle in $BA_2(\mathbb{Z}_{(6)})$.

$$[e_1, e_1 + e_2] + [e_1 + e_2, 4e_1 + 3e_2] + [4e_1 + 3e_2, e_1 + 5e_2 + 2] + [e_1 + 5e_2, e_1]$$

Its reduction modulo 6 inside $BA_2(\mathbb{Z}/6)$ then is a cycle which one can check loops around the torus once; hence it pairs nontrivially with some cocycle. The pullback of this cocycle then by adjunction fails to annihilate the cycle in $BA_2(\mathbb{Z}_{(6)})$, showing it is nonzero in the H_1 . (This example was found with the aid of Claude Opus 5.)

This gives a negative answer to [KMPW, Question 5.1], in that this is a Euclidean domain (as are all semilocal PIDs, using the sum of the valuations) generated additively by its units, which fails to have a Bykovskii presentation.

REFERENCES

- [CP] Thomas Church and Andrew Putman. The codimension-one cohomology of $SL_n\mathbb{Z}$. *Geometry & Topology* 21.2 (2017), pp. 999–1032.
- [KMPW] Alexander Kupers, Jeremy Miller, Peter Patzt, and Jennifer C. H. Wilson. On the generalized Bykovskii presentation of Steinberg modules. *International Mathematics Research Notices* 2022.13 (2022), pp. 10347–10401.
- [X2] Peter Xu. “Presenting restricted Steinberg modules of general linear groups”. Preprint.